Enhancing Clinical Trust in Deep Learning Models Through Grad-CAM-Based Explainability in Radiology Image Classification

A Critical Review of the Evidence and a Layered Evaluation Framework

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Abstract. A large number of deep learning classifiers have been shown to achieve performance comparable to or even better than that of human experts on a wide array of 2D radiology tasks. However, their lack of explainability has thus far prevented them from being deployed in clinical settings. One of the most popular methods for providing insight into how a deep learning classifier arrived at a particular decision is through the use of Gradient-weighted Class Activation Mapping (Grad-CAM), a method that produces a Heat Map-like region of interest which can be readily audited by a human radiologist. This paper provides a structured critical review of the literature that supports the three claims that are typically grouped together by the field, namely: 1) Grad-CAM localizes pathology in images; 2) Grad-CAM 'explains' how the network arrived at its decision via the heat map it produces; and 3) Grad-CAM produces trust in a classifier that is calibrated to the classifier's accuracy. Instead, this paper finds that while Grad-CAM localizes pathology in images somewhat better than chance, there is only weak evidence to support claim 1; that instead there is strong evidence to refute claim 2 for a number of very common deep learning network topologies; and that claim 3 is true only conditionally. The saliency of a given image region by a deep learning network is found to be weakly correlated with its actual relevance to a network's classification of that image. Instead, benchmarking studies of saliency methods find that localization by any method studied here is far below that of human experts in a number of tasks. A series of randomization tests of importance values produced by a variety of state-of-the-art saliery methods found very little to no variation in the importance values for a given image region as the weights of a network were repeatedly and randomly reinitialized and trained on the same image. Relatedly, studies of the use of saliery methods to provide explainable feedback to human users found that such methods only increased accuracy when the underlying model was correct, but decreased accuracy when the model was incorrect and the user relied on the explanation provided by the saliery method. To address these open research questions, this paper provides a novel four-layer framework for the evaluation of a spatial explanation, i.e., an explanation produced for a classifier that is also spatially localized within an image, and distinguishes that from existing frameworks for the evaluation of general-purpose explanation methods. The framework consists of computational faithfulness, anatomical correctness, decision utility, and workflow viability. Relatedly, seven research gaps are outlined and correspond to seven testable study designs. In sum, explanation quality should become a primary endpoint to be reported by researchers studying the performance of deep learning-based image classification algorithms, in addition to more traditional measures of performance such as classification accuracy.

Keywords: explainable artificial intelligence; Grad-CAM; radiology image classification; saliency map faithfulness; trust calibration; human-centered evaluation

1. Introduction

Recent systematic reviews provide evidence that performance of deep learning based automated radiology interpretation has reached a new performance ceiling. Classification accuracies in the high

nineties have been reported on various curated benchmarks for brain MRI, chest radiography and CT scans in controlled setups [1, 2].There is a significant difference between reported performance numbers of automated radiology interpretation systems in deep learning studies and the actual performance in routine clinical practice. This limitation is mostly due to the lack of transparency and interpretability of the decision making process rather than the limited classification accuracy [1, 2].

The main driver for this requirement is structural not cultural. A person who produces radiology reports for a living is professionally accountable for those reports. They will not sign a report for a prediction that cannot be interrogated. Other reviews of medical imaging AI have highlighted the need for medical automated diagnostic systems to be transparent, accountable and to have been aligned with relevant regulations prior to deployment [2]. The research area of explainable AI (XAI) is currently focused on trying to meet these requirements.

The term trust is often used very loosely in the applied XAI papers, and peeling away the layers of approbation to get to the central question of what they really mean by this term, is to provide a definition of trust following the human factors treatment of trust in automation, that is an agent's attitude towards using it to achieve goals in situations of uncertainty. The corresponding terms for reliance on and calibration of such trust are reliance on and calibration of trust, respectively, corresponding to the system's capabilities [3].Calibrated trust increases when the system is reliable and decreases when the system is not reliable. Un-calibrated trust on the other hand increases whenever an XAI explanation is displayed, even if the underlying prediction remains incorrect, and leads to overreliance.

To make this clear, we move from the goal of increasing trust of the clinician with the model to making the trust of the clinician with the model correlate with the correctness of the model. Throughout this paper the trust in the model will therefore be evaluated against the measure of the model's correctness and not against the clinician's confidence in the correctness of the model. Within the area of medical imaging, one explanation method gets more usage than all others by far. In the subset of radiological cardiovascular imaging studies, the most frequently reported XAI technique is Grad-CAM [4], followed by SHAP [5], [6]. As noted above, Grad-CAM holds a similar position in the other areas of medical imaging that are reviewed, the XAI methods most frequently reported are LIME [7] and layer-wise relevance propagation [2].This high utility comes at low cost because Grad-CAM has favorable cost, for it needs not a single retraining step, nor the setup of a substitute model (e.g., a 'proxy' model) to be explained, and not even a single round of perturbations (i.e., feature-analyzing) in order to be made understandable. Instead, the Grad-CAM scores simply are computed by a single pass of backward propagation through the last convolutional layer of a given image classification model. A resulting heat map can then be merged with the original study images to arrive at justifyable results, e.g., in the form of images on which a glioma-inspired model learned to focus.

In this paper we reveal an unwarranted assumption that has been made by the health care computing community on the basis of appearance. For long three different claims have been incorrectly simplified and so misused in many application papers, namely the localization claim (the highlighted part of the image is inside the lesion area), the faithfulness claim (the highlighted part of the image is used by the network), and the calibrated trust claim (showing the saliency map to a human clinician increases the human's reliance on the model's predictions and the clinician's reliance is calibrated to the model's reliability on the particular case). As one would expect from three quite different claims, all can be falsified. Each of them has been empirically tested and the results in each case are quite surprising. Moreover, each result has important implications on how to correctly use the saliency map method of model explanation, Grad-CAM, for human care. The general warning that current explanation methods give false assurances to clinicians in health care [8] as well as the specific

assertion that for high-stakes decisions, we need models that are interpretable by design [9] can both be related to this mistaken conflation of three quite different claims.

The contributions of this paper are as follows.

- A structured critical review of Grad-CAM in radiology image classification that separates the technical, quantitative and human-centered strands of evidence rather than merging them into a single narrative of progress.
- An explicit account of why the gradient averaging step in Grad-CAM breaks faithfulness for a defined class of network topologies, and what this implies for the architectures most commonly used in radiology.
- A four-layer evaluation framework separating computational faithfulness, anatomical correctness, decision utility and workflow viability, positioned against existing general-purpose explanation quality taxonomies and specialised to radiology reporting.
- Seven research gaps expressed as testable study designs with named primary endpoints rather than as general calls for further work.

Section 2 outlines Grad-CAM and its extensions in detail. Section 3 describes the methodology adopted in this review. Section 4 outlines current practices in various applications. Section 5 presents related evidence that are pertinent to the dimensions of localization, faithfulness and calibrated trust. Section 6 outlines the evaluation criteria. Section 7 outlines the research agenda. Sections 8 and 9 detail the limitations of this review and concludes..

2. Grad-CAM and Its Variants: Technical Background

2.1 From CAM to Grad-CAM

Class activation mapping was originally designed for architectures that end with global average pooling and a single linear layer to classify the images. There, the localizable representation is exposed and can be used to compute class-specific spatial evidence by multiplying the final feature maps with the corresponding classifier weights [10].

Grad-CAM removes the restriction by deriving the weights from gradients instead of from classifier parameters. Let A denote the activation tensor of a chosen convolutional layer, with A(k) the k-th feature map and A(k, i, j) its value at spatial position (i, j). Let y(c) be the pre-softmax score for class c. Grad-CAM computes a scalar importance weight for each channel by global average pooling of the gradient of the class score with respect to that channel [4]:

$$\alpha(k, c) = (1 / Z) \cdot \Sigma_i \Sigma_j \; \partial y(c) / \partial A(k, i, j) \quad (1)$$

where Z is the number of spatial positions in the map. The explanation is the ReLU rectified weighted combination of the channels:

$$L(c, \text{Grad-CAM}) = ReLU(\Sigma_k \alpha(k, c) \cdot A(k)) \quad (2)$$

This process of rectification retains only parts of the feature map that have a positive influence on the class of interest. The resulting feature map is then upsampled using bilinear interpolation to the input resolution of the image. As we already pointed out in the original class activation mapping paper [10], the spatial resolution of the explanation is fixed by the spatial resolution of the chosen layer. For standard image classification architectures (i.e. for input images of 224x224 pixels) this corresponds to a 7x7 or 14x14 grid. The fine structures within the resulting feature map for class activation mapping therefore are always mere artifacts of the bilinear interpolation and do not reflect any information that

the network itself has computed. Clinical reader studies have over and over again identified this coarseness of the feature maps as a major practical problem [11, 12].

2.2 Why gradient averaging breaks faithfulness

The averaging in (1) is the source of a specific and provable failure. Draelos and Carin showed that for any convolutional network terminating in a single fully connected layer, the element-wise product of gradient and activation, rather than the channel-averaged product, is guaranteed to reflect the locations the model used [13]. Their method, HiResCAM, replaces (2) with a Hadamard product:

$$L(c, HiResCAM) = ReLU(\Sigma_k (\partial y(c) / \partial A(k)) \odot A(k)) \quad (3)$$

In contrast, Grad-CAM does not guarantee anything. A single scalar weight is applied uniformly to all pixels in a channel, so a location with very small or even opposite contributions can receive high importance only because it is active in a channel where the average gradient is large. Consequently, rather than introducing random noise, the explanations are systematically distorted: Grad-CAM explanations are, on average, larger, more rounded, and smoother than the computation actually supports [13]. While in natural images this might look only slightly worse than the original computation, in radiology images a few millimeters of shift in a highlighted region can change the implicated anatomical structure significantly.

There is an important qualification to this. The HiResCAM map is identical to the corresponding Grad-CAM map when the chosen layer is immediately preceded by global average pooling (e.g. as in ResNet, DenseNet). However for architectures that follow a layer with fully connected layers (and possibly associated flattening operations) or other processing, there is non trivial distortion of the corresponding Grad-CAM map. In other words, the appropriate notion of faithfulness is architecture-conditioned and is not commonly reported in applied papers.

2.3 The variant landscape

Several families of approaches have subsequently been developed to extend and improve the original Grad-CAM formulation, including for example Grad-CAM++, that replaces the uniform channel weighting with pixel-wise weighting of the second order derivatives of the predicted class score [14]. Here the positive gradient contribution at position (i, j) is used as the numerator for a normalized second order term with corresponding weights given by

$$\alpha(k, c, i, j) = [\partial^2 y(c)/\partial A(k,i,j)^2] / [2 \cdot \partial^2 y(c)/\partial A(k,i,j)^2 + \Sigma_a \Sigma_\beta A(k,a,b) \cdot \partial^3 y(c)/\partial A(k,i,j)^3] \quad (4)$$

Score-CAM abandons gradients entirely. Each activation map is upsampled, normalised and used to mask the input, and the channel weight is the resulting increase in the class score relative to a baseline input [15]:

$$w(k, c) = f(X \odot s(Up(A(k))))(c) - f(X_0)(c) \quad (5)$$

The CamupleXD cost function reads as follows: cost = { forward(X_up)^s, X, X0 } where Up denotes upsampling to input size, s a normalisation to the unit interval, X the input and X0 a baseline. To remove gradient saturation and noise, one forward pass per activation channel is required. As a result, the number of network evaluations for an explanation increases with the channel count of the target layer, and thus forms the binding cost for volumetric explanations. LayerCAM extends class activation mapping to earlier layers in the network by using the positive location-specific gradients to weight the activations [16]. The trade space is summarized in Table 1.

Table 1. Class activation mapping methods and their relevance to radiology classification.

Method	Channel weighting	Property gained	Cost or limitation in radiology
Grad-CAM [4]	Global average of first order gradients	Single backward pass, architecture agnostic within CNNs	Coarse maps; no faithfulness guarantee when the target layer is not followed directly by global average pooling
Grad-CAM++ [14]	Pixel-wise weights from higher order derivatives	Better handling of multiple lesions in one field of view	Retains the resolution ceiling of the target layer; still gradient dependent
Score-CAM [15]	Confidence increase under masked forward passes	Gradient free; not affected by gradient saturation or noise	One forward pass per activation channel, so cost scales with channel count and becomes limiting for volumetric CT and MRI
HiResCAM [13]	Element-wise gradient and activation product	Provable faithfulness for CNNs ending in one fully connected layer	Equivalent to Grad-CAM when the target layer precedes global average pooling, so the gain is architecture dependent
LayerCAM [16]	Positive location-specific gradients at shallow layers	Higher spatial detail for small structures	Shallow layers carry less class-specific semantics, so maps can become noisy

2.4 The transformer assumption

Grad-CAM was developed for convolutional feature hierarchies where spatial structure is preserved through the network. Thus, while Vision Transformers and hybrid convolutional-transformer models are increasingly reported for applications in chest radiograph and MRI classification [2], they do not satisfy the same premises. The tokens in a Vision Transformer are not spatially local as are features in a convolutional feature hierarchy, the residual streams in a Vision Transformer generally merge information from different parts of the image at each block, and layer normalization of the residual streams prior to the classification head of a transformer-based model alters the gradient geometry of the feature maps. Thus, while Grad-CAM can be computed on a reshaped token grid and often produces seemingly logical feature importance maps, no faithfulness result equivalent to that for convolutional feature hierarchies has been established for a transformer-based model. It remains an untested assumption of papers in vision that replace a ResNet backbone with a transformer and apply Grad-CAM in the same manner to the resulting model.

3. Review Method

This work is a structured critical review rather than a systematic review conducted under PRISMA, and it is reported as such. The objective was not to enumerate every application of Grad-CAM in radiology, which recent systematic reviews have already done at scale [2], [5], but to assemble the strongest available evidence bearing on the three claims identified in Section 1.2 and to determine whether they are jointly supported.

The sources used to develop this paper were found using IEEE Xplore, Scopus, PubMed and PubMed Central, SpringerLink, and arXiv.org. The search terms were entered into these databases in February 2026 to identify relevant articles that were published after 2016, the year that class activation mapping was introduced. Given that the human factors material used to establish the material in Section 1.1 was developed prior to 2016, that material was not subject to the same time constraints as the rest of the paper. The search terms were combinations of the names of the explanation methods (e.g. Grad-CAM, Grad-CAM++, Score-CAM, HiResCAM, saliency map, class activation mapping) and terms that

denoted material related to radiology (e.g. radiology, chest radiograph, CT, MRI, ultrasound), and terms related to the evaluation of the methods (e.g. faithfulness, localization, trust, reader study, clinician evaluation).

There are three key criteria which we applied in our inclusion of methodological work, evaluations and human-centered studies: 1) Do they tell us something about a formal property of CAM? Then, these methodological works were included regardless of domain. 2) Are there quantitative evaluations including an external reference standard (e.g. an expert segmentation, or a control condition where images were randomly permuted)? We prioritized these over descriptions of what example overlays had been produced by previous authors. 3) Are these human-centered studies with a controlled design and a measurable end point, as opposed to (for example) surveys of people's attitudes towards something? We prioritized these over many other types of work. Preference was given to open access sources so that the evidence that we present here can be directly inspected by the reader; the majority of the works that we cite are available under open licenses.

The evidential core consists of 23 studies, i.e., 7 method papers on CAM and its correction, 5 quantitative studies on the evaluation of XAI methods, 3 controlled studies with a human-centered approach, 4 systematic/narrative reviews, 3 studies on explanation and trust (positions/surveys), and 1 study on human factors of trust in automation. Although many more studies on interpretability present a small set of heat maps on top of an accuracy table, these studies are not included in the core as they only provide evidence of current practice; these are then discussed in Sect. 4. Note that more than three hundred XAI method papers evaluate their method only on the basis of anecdotal evidence [17].

4. Current Practice in Radiology Image Classification

4.1 Architectures, modalities and reporting patterns

Most current work involves fine-tuning a pre-trained (or even transfer-learned) convolutional backbone such as part of EfficientNet, ResNet, DenseNet, VGG, or Xception on a "dataset of interest" (public or from the institution where the work is done), and then presenting classification metrics for the new task. Qualitative support for the XAI method is typically provided by displaying Grad-CAM overlaid images for a few cases where the model got the classification right. Systematic reviews of deep learning in particular areas (e.g. radiological cardiovascular imaging) confirm the method's dominance, but also expose the so far largely qualitative and thus non-standardized nature of the method's evaluation [5]. A more recent and comprehensive review of XAI methods for medical imaging in general (not restricted to any modality) similarly concludes that their current evaluation is for the most part qualitative and thus lacking in standardization [2].

This pattern can also be found in other modalities. For instance, in brain MRI, Grad-CAM is used to validate the focus of the tumour type classification network on the lesion and not on the surrounding parenchyma or even on acquisition artifacts. In chest radiography images, saliency map-based methods, such as Grad-CAM, are used for pneumonia, COVID-19, and pneumothorax classification, where the target findings can be mainly focal, but also can be distributed all over the image in a more diffuse manner. In computed tomography images, Grad-CAM is used to improve the performance of lung nodule and other lesion classification. Due to the volumetric nature of CT images, such methods are either slice-wise, i.e., neglecting inter-slice context, or 3D convolutional, in which case a severe interpolation step is required to map a coarse activation volume back to the input volume. Finally, Grad-CAM has also been used for breast lesion characterization from ultrasound images. Such images are typically acquired in an operator-dependent manner, thus introducing large amounts of variability that cannot be captured by saliency map-based methods.

1. The selection of the displayed cases is almost never specified. The chosen overlays are selected for being good examples of a method's functioning. Readers have no idea how a method will deal with the difficult and the misclassified cases.

2. The accuracy of a method and the generated heat maps (or other explanations) are not related to each other in any statistical way. The heat maps of a method could for example also be generated by a randomly initialized network. Therefore they would look exactly the same when used in a standard paper. This is the exact control that randomization tests are designed for [18].

4.2 Hybrid and multimodal explanation pipelines

Recent work attempts to move beyond the simple heat map of a heatmap by pairing the spatial information provided by Grad-CAM with additional explanation modalities. Often, this SHAP [6] is used to provide feature-level attribution for structured clinical data in addition to providing a spatial attribution for the image, thereby enabling the resulting explanation to state where in the image the model looked as well as which of the patient's characteristics affected the model's prediction. Other approaches have been to combine the spatial information provided by the Grad-CAM heat maps with generated natural language. For example, this natural language can be generated by using a language model to translate the evidence found in the region of interest to natural language that can be read and acted upon by a clinician in the context of a radiology report. Finally, there has been recent work that introduces counterfactual explanation to the clinical domain. Here, rather than simply pointing to areas of the image that the model used to make a classification decision, the approach provides the minimum amount of change required to the image to cause the model to make a different classification decision. This is a question that salierry alone cannot answer.

While these additional pipelines do increase the types of explanations that can be generated and thus are more rich, they suffer from the same fundamental problems as before. Thus a natural language explanation of an unfaithful spatial component is merely a fluent restatement of the already unfaithful claim, possibly more persuasive than a heat map of the errors in the spatial component. As noted above, combining different explanation modes for a component for which there are no validity guarantees for that component can increase the overall persuasive power of the system's output faster than it increases the system's evidence support [8, 9].

5. What the Evidence Establishes

5.1 Localization performance against a reference standard

To evaluate localization, we compare saliency-derived segmentations to expert-annotated segmentations of diseased regions of interest in chest radiographs for a fixed set of pathologies (e.g. pneumothorax, pulmonary edema, etc.). This has been done at scale for the case of chest radiograph interpretation in [19], where seven saliency methods are compared on seven different architectures, and a human benchmark for multilabel segmentation of chest radiographs is established. The results show that saliency localization is significantly and substantially worse than the expert-annotated segmentations for both mean intersection over union and hit rate measures. Importantly, the performance gap with the human benchmark did not correlate with the AUROC of the underlying classification model for any of the pathologies. In particular, for seven out of ten pathologies, the hit rate of the saliency pipeline was significantly lower than that of the human benchmark. The largest performance gap with the human benchmark was observed for the following characteristics of pathologies: multiple instances, large extent of affected anatomy, and high complexity of the affected structure's shape. This is precisely the set of characteristics for which the reader would most like to use localization support.The decoupling of classification accuracy from localization quality in that

study deserves emphasis. It means that a model that classifies well provides no assurance that its heat maps localize well, and therefore that reporting accuracy as evidence of explanation quality is a category error.

In a related work [20], Arun et al. evaluate various set of properties (i.e., localization utility, model weight randomization sensitivity, model repeatability, architecture-level reproducibility) on two different medical image datasets, i.e., SIIM-ACR pneumothorax and RSNA pneumonia datasets. In terms of localization utility, most of methods for visualizing classifiers' saliency are outperformed by a simple localization network (detection/segmentation model) even when the latter is trained and tested on the same data as the former. Moreover, most methods have low repeatability and architecture-level reproducibility. Hence, for tasks where localization is the end task, it is better to train a detection/segmentation model to directly output a localization rather than to rely on saliency methods to extract gradients from a classifier.

Just because quantitative results are reported, that doesn't mean all results were negative. For example, using a statistical attribution framework on a set of chest radiography and brain MRI cases, Brima and Atemkeng found that Score-CAM, XRAI, Grad-CAM and Grad-CAM++ generated"Well-localized attribution maps that targeted relevant anatomical regions of interest"and that "quantitative methods to analyze the distributions of attribution scores provide complementary insights that are not apparent by simple visualization of scores"[21]. Thus, while Grad-CAM often points to relevant features, it is not sufficiently accurate or stable to be used for individual clinical decisions.

5.2 Faithfulness and architectural sensitivity

Evidence of faithfulness is more damaging than evidence of localization because it shows that the explanation is not even describing the model. The standard method for testing faithfulness using randomization tests was introduced by Adebayo et al. [18]. Many commonly used methods do not pass this test. Recently, this type of evidence was also found to have limited sensitivity to weight randomization in a medical imaging replication study on a large number of radiology tasks using a variety of saliency methods, with only limited sensitivity found for several of the methods [20].

The architectural dimension is largely unexplored, however. In their work to systematically evaluate Grad-CAM of varying network depths and architectures for automatic pneumothorax diagnosis, Qiu, Rivaz and Xiao found that increasing network depth does not necessarily improve diagnostic accuracy and that the effectiveness of Grad-CAM further varies with architecture [22]. Thus, explanation reliability is a joint property of a method and its corresponding backbone. However, backbone selection in applied papers on radiology is typically driven by the classification performance on a given task and the availability of corresponding pre-trained models, and thus has never been audited for its explanation reliability.

Three findings from Section 2.2 are particularly relevant when taken together: (1) Grad-CAM has no faithfulness guarantee for a defined class of topologies [13]; (2) Grad-CAM is often insufficiently sensitive to changes in the weights of a network [18], [20]; and (3) Grad-CAM behaves differently at different depths in a network and is even architecture dependent in ways that are not correlated with accuracy [22]. None of these limitations are inherent to the technique as originally proposed for coarse localization by the method's originators. Rather, they are limitations that currently hinder the use of Grad-CAM in clinical research.

5.3 Human-centered evidence

The trust claim can only be tested on humans, and the controlled studies that have done so return a more complicated result than the applied literature assumes.

The human-centered evaluation study of Grad-CAM and LIME by Ihongbe et al. included clinical participants for the evaluation of both models on scenarios of pneumonia and COVID-19 using chest radiographs and CT images [11]. While Grad-CAM was found to be preferred by the participants, they reported specific issues with the clinical usability of the models. Importantly, while the preference of Grad-CAM over LIME is often reported, the specific reservations related to clinical usability that were raised by the participants are hardly ever reported. Importantly, the study concludes that the best people to define the needs of imaging output are the clinicians and therefore the development of models for clinical decision support should be directed to their needs.

Prinster and colleagues conducted a controlled study where radiologists and non-radiologist physicians viewed and interpreted chest radiographs under the guidance of and with the assistance of AI in the form of simulated advice. In this work the advice was varied in terms of the explanation provided as well as the model confidence stated by the AI [23].There was a significant interaction effect for the model's correctness such that when the model was correct, local feature-based explanations resulted in higher mean diagnostic accuracy than global prototype-based explanations (92.8% vs. 85.3%). However, when the model provided incorrect advice, the accuracy for both local and global explanations plummeted to 23.6% and 26.1%, respectively, and there was no significant difference between the two explanation types. Notably, local explanations facilitated alignment with the model's advice more quickly than global explanations, whether or not the model's advice was accurate. Thus, local explanations improved performance when the model was already correct but facilitated uncalibrated trust when the model was not, a finding that underscores a key point outlined in Section 1.1.

These results are particularly powerful for radiography, as the study investigated the saliency of heatmaps and binary model output with confidence from skeletal radiographs by student and qualified diagnostic radiographers [12]. For both groups, saliency of heatmaps and confidence of model output added value to radiographers' accuracy in reading skeletal radiographs. However, there was a marked reduction in accuracy for pathological cases by student radiographers following presentation of the saliency of heatmaps before returning to baseline following presentation of the binary model decision with confidence. Importantly, the trust that participants reported in the system as a whole declined from start to end of the study. Importantly, this decline in trust reported by all participants was due to the lack of granularity in the heatmaps provided as explanation. In essence, explanations that are not granular enough to be relied upon do not simply fail to add value, they actually decrease accuracy in the group with least experience, and that is the group for which decision support is most commonly offered.

Table 2. Controlled evidence bearing on the localization, faithfulness and calibrated trust claims.

Study	Design	Setting	Finding relevant to trust
Saporta et al. [19]	Benchmark against expert segmentations	Chest radiography, ten pathologies	Saliency localization significantly worse than the human benchmark on mIoU and hit rate, independent of classification AUROC
Arun et al. [20]	Utility, randomization, repeatability and reproducibility tests	Pneumothorax and pneumonia radiographs	Most methods underperform dedicated localization networks and show weak repeatability and reproducibility
Qiu et al. [22]	Systematic comparison across depths and architectures	Automatic pneumothorax diagnosis	Grad-CAM effectiveness varies with architecture; deeper networks do not reliably improve accuracy

Study	Design	Setting	Finding relevant to trust
Draelos and Carin [13]	Formal proof with empirical illustration	Method level, transferable	Grad-CAM can highlight locations the model did not use; HiResCAM carries a faithfulness guarantee for a defined topology class
Ihongbe et al. [11]	Human-centered user study	Pneumonia and COVID-19, radiographs and CT	Grad-CAM preferred over LIME, but with explicit clinical usability reservations
Prinster et al. [23]	Controlled reader study with simulated AI advice	Chest radiographs, experts and non-experts	Local explanations raised accuracy when advice was correct (92.8 against 85.3 percent) but not when it was incorrect (23.6 against 26.1 percent), and accelerated alignment with the model either way
Rainey et al. [12]	Controlled study of AI feedback formats	Skeletal radiographs, students and qualified radiographers	Heat maps reduced accuracy for pathological cases in the student group; self-reported trust declined over the study

5.4 Synthesis: an explanation and calibration decoupling

Putting all these together, Grad-CAM increases perceived interpretability of a system. But it does not increase localization accuracy over purpose-built methods [19, 20]. There is no guarantee of faithfulness of the Grad-CAM results, and the degree to which they are faithful depends on architecture [13, 22]. The impact on clinician performance is dependent on model correctness, reader expertise, and explanation format, and can lead to increased reliance on the model without corresponding improvement in clinical outcomes on cases where the model fails [12, 23].

A safety-relevant but asymmetric consequence of model explanation is demonstrated in the two controlled studies. An explanation is either helpful or redundant in addition to correct model predictions. However, in case of model errors, apparent evidence for incorrect model predictions is supplied. In contrast, reader accuracy is not rescued by good quality of the explanation. Thus, explanation quality matters where evaluation of learning systems today mainly focuses on correctly classified illustrative cases for which no information is provided in practice. Our results support the general warning of false reassurance of current explainability approaches in health care [8] and support the development of models that are interpretable by design where the task permits this [9].

6. A Four-Layer Evaluation Framework

The disagreements in the literature are in part definitional. Studies reach different conclusions about whether Grad-CAM works because they are testing different propositions under a single word. We therefore propose a framework that separates four questions, each with its own reference standard and its own failure mode. Figure 1 shows the separation of claims from Section 1.2 alongside the layers that test them.

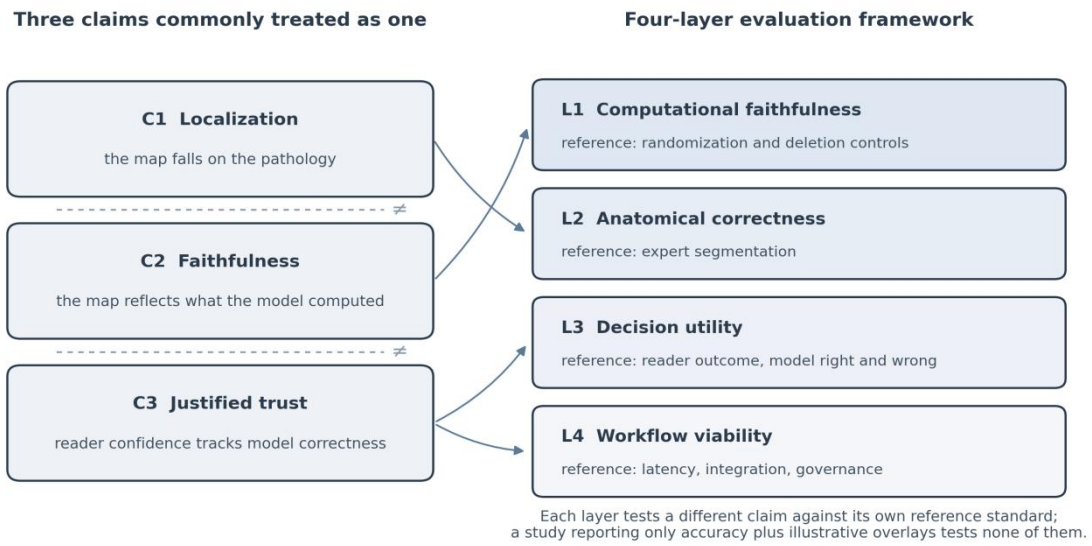


Figure 1. The three claims are logically independent, and each evaluation layer tests a different one against its own reference standard.

6.1 Relation to existing evaluation taxonomies

Note that this framework will not replace general-purpose schemes to evaluate explanation quality. The set of Co-12 properties of explanations as compiled by Nauta and colleagues in [17] provides the most comprehensive conceptual categorization of explanations so far. This set of 12 properties of explanations is complemented by a large collection of quantitative evaluation methods mentioned in more than three hundred papers on XAI [17]. Van der Velden et al. [1] provide a criteria-based classification of explanation methods in medical imaging. This classification is based on anatomical location and type of explanation. As such, it is a descriptive tool that classes what can be measured.

There are three key differences in our framework compared to other taxonomies for studies of radiology explainers and these differences constitute the contribution of this work. First, this is a prescriptive framework, setting out a minimum set of items that should be reported in a study of a radiology explainers, as opposed to a purely descriptive framework. Second, it is anchored on known reference standards in a radiology study, so that each layer of the framework includes a "comparator" which is a known entity in a study of a radiology image such as expert human segmentation, randomization, reader study and deployment. Third, it includes two new layers that are not typically included in general-purpose taxonomies, decision utility (i.e. value) conditioned on model correctness and workflow viability. These are properties of deployed systems not of individual explanations, and are shown in Section 5.3 to be crucial. Co-12 lists many relevant properties but a study can satisfy all of these and still fail to address clinical questions of interest to the authors of this paper because Co-12 does not ask what happens to the reader when the model is wrong.

6.2 The layers

Table 3. Four-layer evaluation framework for explanation quality in radiology classification.

Layer	Question answered	Candidate metrics and controls	Maturity of current evidence

Layer	Question answered	Candidate metrics and controls	Maturity of current evidence
L1 Computational faithfulness	Does the map describe what the network actually computed?	Model and data randomization tests; deletion and insertion curves; comparison against a provably faithful method on the same backbone	Method-level results exist [13], [18] but are almost never reported in applied radiology papers
L2 Anatomical correctness	Does the map fall on the structure a radiologist would identify?	Intersection over union and hit rate against expert segmentation; pointing game; per-pathology stratification by lesion size and multiplicity	Benchmarks exist for chest radiography [19], [20]; sparse for CT, MRI and ultrasound
L3 Decision utility	Does the map improve reader decisions, including when the model errs?	Reader accuracy with and without the map, reported separately for correct and incorrect model advice; speed of alignment with the model; trust calibration against model correctness	Three controlled studies [11], [12], [23], with conditional and partly negative results
L4 Workflow viability	Can the map be produced and consumed in the real reporting environment?	Inference latency at clinical volume; PACS and reporting integration; reading time impact; governance and audit trail	Largely unaddressed; identified as a barrier in human-centered work [11]

We note three key properties of the framework. First, the layers are ordered but not necessarily hierarchical. Thus, a map could be anatomically accurate but not faithful. Each layer has its own reference standard, so it is not possible to condense the quality of the explanations to a single number. Most importantly, a paper containing only the classification metrics of a method and some colorful overlays would not fulfill any of the four layers of the framework [2], [5], [17].

The framework as applied here requires a single metric per layer. So for a study of a DenseNet based classifier for the detection of pneumothorax, a randomization control is measured at L1, a metric of the intersection over union (IoU) of a method's segmentation against masks created by an expert is measured at L2 stratified by lesion size, a small reader study is measured at L3 where accuracy is reported separately for cases in which the model correctly or incorrectly advised the reader, and the per-study inference time is measured at L4. None of these is particularly difficult to measure, their absence is simply a convention.

7. Research Gaps and a Testable Agenda

The following gaps are stated with the study design that would close them, because the recurring weakness of research agendas in this area is that they call for more work without specifying what would count as an answer.

7.1 Gap 1: Absence of prospective in-clinic validation

All evidence on Grad-CAM in radiology to date has been retrospective and offline, including controlled reader studies that were simulated. A critical unknown is how image explainations affect the decisions made by radiologists operating under time pressure in real world environments where there are real consequences for their decisions. A prospective study is required with a randomized design at the level of the reading session. A single arm would compare the reporting of cases using the model output to that where the radiologist has also been provided with explainations of the model output. The primary endpoint would be diagnostic accuracy measured against a delayed reference standard. Reporting time, number of discrepancies and instances of deferral would also be measured as

secondary endpoints. Results on those cases where the model was wrong should be presented separately and as pre-specified. The nature of the aggregate estimate for these cases is misleading, as outlined in Section 5.3.

7.2 Gap 2: Architecture-conditioned explanation reliability

The reliability of explanations in terms of depth and architecture is not fully understood [22] and a faithfulness guarantee can only be provided for a certain section of the network following a target layer [13]. Unfortunately, no such systematic map exists for medical imaging tasks and architectures. For this reason, a study is required that fixes a task and corresponding dataset, but varies the backbone family, corresponding depth and the structure of the used head. The results should be presented as a table with corresponding L1 and L2 metrics for each architecture. Such a table allows for the selection of a suitable backbone for explanation purposes, in addition to classification accuracy.

7.3 Gap 3: No agreed explanation quality metric set for radiology

Many studies on current reporting use incompatible endpoints. Even a general catalogue of quantitative methods of evaluation already exists [17]. What is missing and required is a minimum of agreed methods for the evaluation of radiological images. A simple metric for each layer of evaluation as outlined in section 6 could be sufficient. The creation of such a minimum of methods for evaluation of radiological images by a Delphi process of stakeholders from radiology, machine learning and government would be sufficient. Without such a minimum of methods for evaluation of radiological images, systematic reviews on evaluation of reporting for computer-assisted detection of tumors in mammography until now only state that the evaluation so far has been qualitative and unstandardized [5].

7.4 Gap 4: Faithfulness by construction rather than post hoc repair

Most variant studies on explainability of deep learning classifiers for radiology have focused on fine-tuning an already post-hoc approximated black box. As alternative, one can design evidence-exposing classifiers by design: prototype-based reasoning, attention that is constrained to be interpretable by design of the architecture, or concept bottlenecks with clinical intermediate variables. A comparison of an interpretable classifier versus a black box that is explained post-hoc for the same task and with same accuracy in all four layers of the framework is not yet done at scale for the specific domain of radiology. While an accuracy penalty for interpretability is a frequent assumption [9], this needs to be measured, not assumed.

7.5 Gap 5: Multimodal explanation with a validated visual component

The task of combining spatial attribution with clinical data and report text is promising but thus far largely unexplored, and likely to be prone to fluent unfaithfulness. To establish the validity of such an explanation, first validate each component against appropriate reference standards, then compare reader calibration on the fused explanation to that of the best single component. To assess the extent to which the narrative component masks degradation of the visual component, include a condition in which the visual component is degraded intentionally.

7.6 Gap 6: Trust calibration and overreliance

Trust that is increased by explanation presence but does not track model accuracy is not an improvement, it is a hazard. We now have evidence that alignment with the model is accelerated when it gives incorrect advice [23] as well as evidence of a drop in accuracy when less experienced readers are presented with models' outputs in the form of heat maps [12].To study the desirability of these kinds of studies (i.e., studies that measure calibration) in more detail, the cases in which the model

generated correct answers and cases in which it generated wrong answers are to be presented to the readers in known and prespecified proportions, and the relationship between the readers' confidence and the model's correctness is to be estimated. As previously observed, the effects here are to differ by the readers' level of expertise, so the variable representing the readers' level of expertise is to be used as a stratification variable and not as a covariate.

7.7 Gap 7: Resolution, small lesions and distribution shift

First, the limit on the resolution of last-layer methods, as described in Section 2.1, affects the hardest cases of small lesions, early findings, and diffuse disease. In order to investigate the ceiling on resolution of last-layer methods for these hardest cases in more detail, two areas of work need to be done. First, the performance of higher-resolution methods such as LayerCAM [16] and Score-CAM [15] must be evaluated in detail for the small-lesion case, and not only on easy cases. Second, the degradation of the quality of the explanations under distribution shift to other scanners, to other acquisition protocols, and to different demographic subgroups must be characterized in detail, since a map that is confident while the rest of the model is degrading silently is a specific and so far poorly understood failure case.

Table 4. Research gaps mapped to study designs and primary endpoints.

Gap	Study design that would resolve it	Primary endpoint
G1 Prospective validation	Prospective study randomized by reading session, model output alone versus model output plus explanation	Diagnostic accuracy against delayed reference standard, with prespecified separate reporting on model-error cases
G2 Architecture dependence	Factorial comparison across backbone family, depth and head structure with task and data held fixed	Faithfulness and localization metrics per architecture and layer combination
G3 Metric standardisation	Delphi consensus with radiologists, researchers and regulators, followed by stability validation	An agreed radiology-specific minimum reporting set with demonstrated inter-rater and inter-dataset stability
G4 Interpretability by construction	Matched comparison of an inherently interpretable classifier against a post hoc explained black box	Accuracy difference alongside all four evaluation layers
G5 Multimodal validity	Component-wise validation before fusion, with a deliberately degraded visual condition as control	Reader calibration relative to the best single component; detection rate of degraded input
G6 Trust calibration	Controlled reading with a prespecified proportion of model errors, stratified by reader expertise	Association between reader confidence and model correctness
G7 Resolution and shift	Evaluation restricted to small and diffuse findings, plus cross-scanner and cross-demographic testing	Localization performance in the small-lesion regime; degradation of explanation quality under shift

8. Limitations of This Review

This is a structured critical review, not a systematic review conducted under PRISMA. Search dates and databases are reported in Section 3, but screening counts and a formal risk of bias assessment are not, and source selection prioritised evidential weight over exhaustiveness. The resulting corpus is deliberately weighted toward methodological and controlled studies, and broad enumeration of applications is available elsewhere [2], [5].

All studies considered in this review were written in English. Preference was given to open access publications to limit the review to work that could be readily accessed, even when published in other

languages or behind paywalls. Given that studies included in this review reported somewhat different endpoints, no attempt was made to combine results quantitatively, and the resulting synthesis is largely narrative in form and therefore open to some degree of interpretation. The evidence base regarding human-centered applications and their effects consists of very few studies in absolute terms. Of those considered in Section 5.3, three were quite different from one another with respect to modality, reader, and explanation form. Two used simulations rather than advice generated from deployed models; and one compared local to global explanations rather than specifically to Grad-CAM-generated explanations. Hence, it would be unwarranted to conclude that results obtained with these studies also apply to Grad-CAM-generated explanations, but rather that they can be taken as indicative of results for spatial explanations more broadly.

Section 6 outlines a new framework. As this is a conceptual contribution, this has not yet been empirically tested or validated. While the layers outlined in Section 6 draw upon documented failure modes, the framework is positioned against currently available taxonomies in Section 6.1. It is however an area for potential future research, in particular to determine whether the minimum reporting set of information outlined is sufficient to distinguish between useful and less useful information for decision-making (i.e. Gap 3).

9. Conclusion

This review builds on the prior work of Grad-CAM, in terms of enabling the inspection of model behavior at little cost above and beyond regular use of the model, revealing instances where models focus on artifacts and confounders rather than meaningful features, and equipping the clinician with terminology to inspect (or debug) otherwise black box systems. However, the rest of the field of saliency methods for deep learning in medicine has proceeded under the mistaken claim that is sustained by none of the evidence.We found that 1) the localization of saliency for deep learning models in general tends to perform below human level and is not strongly correlated with model classification accuracy [19, 20]; 2) that faithfulness of explanations is not guaranteed and tends to vary by model architecture in often unreported ways [13, 18, 22]; 3) that spatial explanation improves human reader performance on specific tasks for correct model predictions, but accelerates shift to relying on models for incorrect predictions [12, 23].Explanation that increases reliance without assessing correctness shifts risk rather than reducing it.

There is a change in reporting practice (as opposed to disuse) and explanation quality is treated as a main outcome variable using a reference standard for the four levels of explanation as outlined in Section 6. In particular, results for the difficult cases and misclassified cases are to be stratified from the results for the illustrative cases. Localization in particular requires localization models [20].The direct comparison with the architecture that builds interpretability into it is overdue [9]. As with any other aspect of a clinical system, trust in it is earned by validated performance under the conditions of use, and the explanation can merely be a means to calibrating such trust, not a replacement for it.

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